

Retail Market Behavior Analysis Using Multi-Task Data Mining Techniques

Progyan Sen, Sumit Nayak, Dev Singh, Kshitij Saxena
Team Name: Miners

Abstract—This document presents a multi-task data mining study of retail market behavior using the UCI Online Retail dataset. The project integrates three core components: (1) customer segmentation using the RFM (Recency, Frequency, Monetary) framework and K-Means clustering, (2) product similarity search using K-Nearest Neighbors (KNN), and (3) market basket and temporal purchasing pattern analysis based on Apriori frequent itemset mining and time-series aggregation. The dataset contains over half a million invoice-level records from a UK-based online retailer, including information about customers, products, and transaction times.

Our analysis reveals three distinct customer segments ranging from low-value casual buyers to extremely high-value customers. KNN-based similarity mapping uncovers meaningful product-to-product relationships, while Apriori discovers frequently co-purchased item sets that can support cross-selling strategies. Time-series analytics highlight peak purchasing hours and the most profitable weekdays. An interactive deployed dashboard¹ demonstrates all modules of the system.

Index Terms—Data mining, retail analytics, RFM segmentation, Apriori, KNN, time-series analysis.

I. INTRODUCTION

Retailers increasingly rely on data-driven methods to understand customer behavior and optimize business decisions. Online retail platforms generate large volumes of transactional data that capture which products are purchased, when transactions occur, and how much customers spend. Mining this data can reveal actionable patterns for segmentation, recommendation, and sales optimization.

In this project, we analyze the *Online Retail* dataset from the UCI Machine Learning Repository, which records transactions from 01/12/2010 to 09/12/2011 for a UK-based online retailer. The company sells unique all-occasion gifts to customers in multiple countries. Using this dataset, our goal is to build a multi-task analytics pipeline that answers key business questions about customer value, product similarity, and purchasing patterns.

The main objectives are:

- Segment customers using RFM-based K-Means clustering.
- Identify similar products using KNN for item-item recommendation.
- Discover frequent itemsets using the Apriori algorithm.
- Analyze time-based purchasing behavior using hourly and weekday sales aggregation.

A web application is deployed to make these analyses interactive and visually interpretable.

¹<https://retail-market-behaviour.vercel.app/>

II. RELATED WORK

RFM (Recency, Frequency, Monetary) analysis is a classical technique in customer relationship management, frequently combined with clustering algorithms such as K-Means for customer segmentation [?]. It provides interpretable dimensions for characterizing customer value and behavior.

Market basket analysis is traditionally performed using Apriori-based association rule mining, which discovers frequent itemsets and strong rules based on support, confidence, and lift [?]. These methods have been widely used to design promotions and recommendation strategies.

Similarity-based recommendation systems often employ KNN-style methods, where neighbors are computed using distance in a feature space derived from product attributes or co-purchase behavior [?].

Time-series analysis has been used to understand seasonality and temporal patterns in retail sales, helping retailers identify peak hours and high-revenue days [?]. Our work follows this line by focusing on pattern discovery rather than forecasting.

III. METHODOLOGY

A. Dataset Description

The UCI Online Retail dataset contains 541,909 transaction records and 8 attributes:

- **InvoiceNo**: Transaction identifier.
- **StockCode**: Product code.
- **Description**: Product description.
- **Quantity**: Number of units purchased.
- **InvoiceDate**: Date and time of the transaction.
- **UnitPrice**: Price per unit.
- **CustomerID**: Unique customer identifier.
- **Country**: Customer location.

After cleaning invalid entries, we retain several thousand customers and over 3,900 unique products.

B. Preprocessing

Preprocessing includes:

- Removing missing **CustomerID** values.
- Removing cancelled invoices (InvoiceNo starting with “C”).
- Filtering non-positive quantities and prices.
- Computing **TotalPrice = Quantity × UnitPrice**.
- Extracting **hour** and **weekday** from InvoiceDate.
- Standardizing RFM values for clustering.

C. Feature Engineering

RFM Features:

- Recency (days since last purchase)
- Frequency (total number of invoices)
- Monetary (total spending)

Product Features for KNN:

- Textual product descriptions converted to embedding vectors.
- KNN retrieves nearest-neighbor product matches.

Time-Series Features:

- Total revenue by hour.
- Total revenue by weekday.

D. Models Used

K-Means ($k = 3$) for RFM Segmentation Used to identify groups of customers with similar purchasing behaviors.

KNN for Product Similarity Used as an unsupervised recommendation system for related items.

Apriori Algorithm Used for discovering frequent co-purchased product combinations.

Time-Series Analytics Only aggregation-based (no forecasting model used).

E. Training and Validation

- Clustering validated through cluster interpretability.
- KNN checked via meaningfulness of nearest neighbor matches.
- Apriori validated using support and domain consistency.
- Time-series validated via aggregated sales trends.

IV. RESULTS

A. Quantitative Results

TABLE I
CUSTOMER SEGMENTATION SUMMARY

Cluster	Customers	AvgSpend	AvgOrders	AvgItems
0	2874	64.61	2.38	38.4
1	14	3037.89	10.86	2159.07
2	5	1783.78	79.00	1153.40

KNN Similarity Examples:

- *HANGING JAM JAR T-LIGHT HOLDER* → similar to *HANGING HEART JAR T-LIGHT HOLDER*.
- *GOLD FISHING GNOME* → similar to *SILVER FISHING GNOME*.

Apriori Market Basket Patterns:

- Frequent combinations among tissue packs, Christmas items, and retrospot products.

Time-Series Insights:

- Best Hour: **12 PM**
- Best Day: **Thursday**

V. DISCUSSION

The results show that even simple models uncover meaningful behavioral patterns. RFM segmentation isolates low-, mid-, and extremely high-value customers. KNN similarity captures product-level relationships useful for recommendations. Apriori mining reveals strong co-purchase themes. Time-series analytics provide actionable insights about peak sales periods.

Limitations include missing demographic context, seasonal variation, and absence of forecasting models.

VI. CONCLUSION

This project integrates clustering, similarity search, frequent itemset mining, and time-based analytics into a unified retail market behavior analysis system. Insights derived from this multi-task approach can support marketing, recommendation, and operational decision-making. Future work includes advanced embeddings, forecasting models, and deeper basket optimization.

REFERENCES

APPENDIX

Full source code is available at:

<https://github.com/Sumit210106/retail-market-behaviour>

A. Repository Structure

- **backend/**
 - **data/** – raw datasets
 - **ml/**
 - * kmeans.py
 - * knn.py
 - * apriori.py
 - * timeseries.py
 - * pca.py
 - * dbscan.py
 - * decision_tree.py
 - **utils/**
 - **app.py** – FastAPI backend
 - **requirements.txt**
 - **idea.txt**
- **frontend/** – Next.js dashboard
- **.github/workflows/** – CI/CD automation
- **.gitignore**, **.DS_Store**

B. Description of Key Components

- RFM-based customer segmentation
- KNN product similarity engine
- Apriori frequent itemset mining
- Time-series revenue analysis
- Interactive dashboard for visualization